

# Khalid Hasan Raj

20/6, North Manikdi, Dhaka Cantonment-1206, Dhaka, Bangladesh

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Google Scholar | ResearchGate

## Current Position

### Lecturer

Department of Biochemistry and Molecular Biology, University of Dhaka

May 2026 – Present

Dhaka, Bangladesh

## Education

### M.S. in Biochemistry and Molecular Biology

University of Dhaka

2025

- GPA: 4.00/4.00 Rank: 1/60

- Thesis: *Harnessing E. coli for cost effective production and purification of a novel pan-serotype multi-epitope vaccine against dengue virus*

- Thesis Supervisor: Professor A.H.M. Nurun Nabi, PhD

### B.S. in Biochemistry and Molecular Biology

University of Dhaka

2023

- CGPA: 3.96/4.00 Rank: 1/60

### H.S.C. in Science

Dhaka College

2018

- GPA: 5.00/5.00

### S.S.C. in Science

Shaheed Ramijuddin High School

2016

- GPA: 5.00/5.00

## Research Experience

### Researcher

Department of Biochemistry and Molecular Biology, University of Dhaka

May 2026 – Present

- Leading multi-omics bioinformatics projects to develop risk prediction models for stomach and bladder cancer.

- Investigating genetic associations underlying non-communicable diseases.

- Collaborating within multidisciplinary research teams, contributing to data analysis, interpretation, and scientific writing.

- Developing research ideas and preparing research proposals and grant applications.

### Research Fellow

International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)

Oct 2025 – May 2026

- Conducted mammalian cell culture and virus culture under biosafety-compliant conditions.

- Performed immunological and virological assays.

- Contributed to biosimilar potency testing and supported clinical trials of viral vaccines.

- Participated in antiviral drug discovery projects.

- Collaborated in multidisciplinary research teams; contributed to data analysis, interpretation, and scientific writing.

#### **Senior Research Assistant**

June 2025 – Sept 2025

*International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b)*

- Maintained mammalian cell lines and assisted in virus culture experiments.
- Supported laboratory operations, data collection, and experimental optimization.
- Assisted in training junior researchers and interns in laboratory techniques and safety practices.

#### **MS Thesis Student**

Sept 2023 – June 2025

*Department of Biochemistry and Molecular Biology, University of Dhaka*

- Designed an immunoinformatics-based multi-epitope vaccine targeting all dengue virus serotypes.
- Cloned the vaccine construct gene into an expression vector and expressed it in *E. coli*.
- Optimized induction conditions and purified the recombinant protein using chromatographic techniques.
- Confirmed protein expression and purity by SDS-PAGE and Western blotting.

### **Teaching Experience**

#### **Lecturer**

May 2026 – Present

*Department of Biochemistry and Molecular Biology, University of Dhaka*

- Conducting undergraduate and graduate courses in Biochemistry, Bioinformatics, Molecular Biology, and Nutrition.
- Designing and delivering lecture materials, tutorials, and laboratory sessions.
- Preparing examination questions and evaluating answer scripts.
- Supervising and mentoring undergraduate students in laboratory-based and computational research projects.
- Advising students on coursework, thesis work, and academic progress.
- Invigilating examinations and participating in departmental academic and curriculum committees.

### **Publications**

1. Raj, K. H., Hossain, E., Zahin, H., Noman, A. A., Saba, A. A., Sayem, M., Yasmin, T., & Nabi, A. H. M. (2025). A robust comprehensive immunoinformatics approach for designing a potential multi-epitope based vaccine against a reiterated monkeypox virus. *Biochemistry and Biophysics Reports*, 43, 102075. <https://doi.org/10.1016/j.bbrep.2025.102075>
2. Akter, S., Fuad, M., Mahmud, Z., Tamanna, S., Sayem, M., Raj, K. H., & Howlader, M. Z. H. (2025). Comprehensive in silico characterization of nonsynonymous SNPs in the human ezrin (EZR) gene and their role in disease pathogenesis. *Biochemistry and Biophysics Reports*, 42, 101972. <https://doi.org/10.1016/j.bbrep.2025.101972>
3. Mowla, A., Mahmud, Z., Raj, K. H., Tamanna, S., Zeba, Z., Ansari, F., & Howlader, M. Z. H. (2026). Genetic association of vitamin D receptor variants and 25-hydroxyvitamin D levels with breast cancer risk and triple-negative phenotype in Bangladesh. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-68260-9>

### **Skills**

**Molecular Biology and Biochemistry:** DNA and RNA extraction, PCR, qRT-PCR, gel electrophoresis, molecular cloning and plasmid handling, heterologous protein expression, SDS-PAGE and western blotting.

**Cell Culture and Virology:** Mammalian cell culture, virus culture, plaque assay, pseudovirus neutralization assay (PNA), focus reduction neutralization test (FRNT), biosimilar potency testing, antiviral drug screening.

**Immunological Techniques:** Enzyme-linked immunosorbent assay (ELISA), neutralization assays, immune response testing.

**Bioinformatics:** Genomics and transcriptomics data analysis, immunoinformatics-based data analysis, protein structure prediction and mutation analysis, molecular docking, molecular dynamics simulation, machine learning model development.

**Programming and Data Analysis:** R (biostatistics, visualization, transcriptomics analysis), Python (data processing, machine learning implementation), Bash (Linux-based data handling and pipeline execution).

**Research and Academic Skills:** Experimental design and optimization, data interpretation and statistical analysis, scientific writing (manuscripts, theses, proposals), research presentation and seminar delivery, training and mentoring junior researchers.

## Awards and Fellowships

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- National Science and Technology (NST) Fellowship – 2024
- Provost Award – 2023, Amar Ekushey Hall, University of Dhaka
- Provost Award – 2022, Amar Ekushey Hall, University of Dhaka
- Dean's Award – 2022, Faculty of Biological Sciences, University of Dhaka
- Dhaka University Honors Scholarship – 2022

## References

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### A.H.M. Nurun Nabi, PhD

*Professor, Department of Biochemistry and Molecular Biology, University of Dhaka*

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### Mohammad Mamun Alam, PhD

*Assistant Scientist, Infectious Disease Division, icddr,b*

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